Weather Conditions and Reported Vehicle Crash Patterns in Selected Lower Mainland Municipalities, 2021-2024

Introduction and Problem Definition

Throughout the years, vehicle crash frequencies have varied across different municipalities and the role of weather in such collisions may affect the statistics and driving behavior which may or may not compromise road safety. This project examines reported vehicle crash patterns in the nearby municipalities of **Vancouver, Richmond, Delta, and Abbotsford** from 2021 to 2024 with significant attention given to finding out if precipitation and temperature are associated with monthly vehicle collision frequencies.

The project combines two data sources, originally in their raw states, which made it difficult for direct comparison. Vehicle crash information was sourced from ICBC's crash open datasets, while the daily historical weather observation datasets were obtained from the Environment and Climate Change Canada (ECCC) website. In order to work clearly with the raw datasets, cleaning and municipality-year-month aggregation was important and needed before any analyses could be performed on the data. In addition to the two datasets mentioned previously, another dataset was obtained from ICBC statistics regarding 'Police-reported crashes' to help us answer some of the questions below, and aid us in analyzing whether directly recorded weather from weather stations and reported road conditions had an underlying yet clear association with reported vehicle collisions.

The analysis addressed five key research questions:

1. How did reported crash counts and casualty outcomes vary across municipalities and years?
2. How did crash frequency vary by months and seasons?
3. Were monthly precipitation and temperature fluctuations associated with crash frequency?
4. Did weather-crash relationships differ among the 4 municipalities?
5. In police-reported crashes, were weather and road conditions associated with accident type and casualty outcomes?

The goal is to identify useful patterns and statistical associations, instead of establishing that weather directly caused changes in crash frequency.

Data Collection and Cleaning

The main ICBC dataset was composed of detailed crash records including, but not limited to; municipality, year, month, crash severity, crash configuration, time information, vulnerable-road-user involvement, total crashes, and total victims. While both the raw ICBC lower mainland dataset and the cleaned version kept the same amount of rows, many columns were removed, specifically those with flags which would only make it harder to translate the data. The best course of action was to remove such column data, while preserving all the rows, however some rows in other datasets contained missing or blank values, which had to be removed or in some cases, it was left alone as outliers to ensure that there isn't a bias in the data. The ICBC dataset for the lower mainland was

covered the 4 cities stated earlier, was cleaned and aggregated by municipality, year, and month to produce “192 monthly observations,” which corresponds to 4 municipalities * 4 years * 12 months.

The second ICBC dataset, “ICBC police-reported” contained 12,230 rows which also kept the same number of rows from the initial raw dataset. This dataset included categorical variables such as accident type, collision type, reported weather, road condition, crash count, casualties, and vehicles involved. The purpose of this dataset was to be used for research question (5) where Chi-Squared test of independence was used given that it includes categorical variables, more on that later.

The ECCC weather observation datasets were collected from 4 different weather stations: **Vancouver Harbour CS, YVR Airport (Richmond), Burns Bog (Delta), and Abbotsford A**. The initial raw datasets for these municipalities were further saved by year, making 16 raw datasets, per city, per year. The result of cleaning and aggregating this dataset resulted in 5,844 rows which match the number of days between these years.

During the data collection, there was a problem with the initial Richmond weather dataset as data collected from “Richmond Nature Park Weather Station” did not have data for 2/12 months of 2023, and the entire 2024. I decided to move these datasets to a folder called deprecated, and then chose to get data from the YVR Airport weather station, which had all the data needed for the yearly timespan. YVR weather station was the closest to Richmond Nature Park, just about 7km away, making the weather datasets almost identical, with a few possible discrepancies here and there.

Daily weather observations were then converted to monthly summaries including mean temperature, total precipitation, measurable precipitation days, freezing days, etc. Missing observations were kept as is rather than imputing the data. The monthly precipitations were only used in precipitation dependent analyses when at least 80% of the month’s daily precipitation observations were available.

The final weather and crash datasets were joined on municipality, year, and month, and our validation confirms that the resulting combined_monthly.csv contained exactly 192 unique municipality-year-month records with no missing or duplicate keys.

Analysis Methods

The analysis combined descriptive and inferential techniques covered in CMPT 353. Pandas and NumPy were used for aggregation and summary calculations, while Matplotlib and Seaborn were used for visualizing data plots and figures. Crash frequency was first summarized by year, month, municipality, and season. Seasons were defined as Winter (December–February), Spring (March–May), Summer (June–August), and Fall (September–November).

Pearson correlation was used to evaluate linear relationships between monthly weather measurements and crash counts, while Spearman correlation was used as a non-parametric comparison that measures monotonic relationships (those that increase or decrease in the same direction). These tests were performed separately for each municipality because the municipalities differ greatly in the overall crash volume. Just to note, the use of Pearson and Spearman Correlation

was not discussed in class, however it was something I stumbled upon during some research, and then Gen AI was used to understand how to use it.

A multiple linear regression model was also fitted using “precipitation”, “mean_temperature”, “municipality”, and “year” to provide a multigraphed view of monthly crash counts. Because the scikit-learn implementation was used, this model was treated primarily as a descriptive supporting model, rather than the main inferential significance test.

To test whether crash distributions differed between relatively wet and dry months, each municipality's months were divided around that municipality's median monthly precipitation. A two sided Mann–Whitney U test was then applied separately to each municipality.

Finally, the police-reported crash data was analyzed with contingency tables and Chi-square tests of independence. These tested whether the accident type was independent of reported weather conditions and road conditions. Rare categories were combined where necessary, and unknown weather or road condition categories were excluded (temporarily removed) from the inferential stats tests. Many of these tests were in the form of helper functions, which were invoked when needed.

Data Analysis Results

4.1 Crash Trends Across Municipalities

The reported crashes were on a steady increasing slope from 2021 to 2024 in all four municipalities. The significance of the bunch was from Vancouver which increased from 40,216 crashes in 2021 to 50,069 crashes in 2024, which translates to about a 24.5% increase over the 4 years. The city of Richmond had the largest proportional increase, going from 12,170 to 16,289 crashes, translating to a 33.8% increase. Delta had an increase of about 16.4%, while Abbotsford had a mere 10.6% increase.

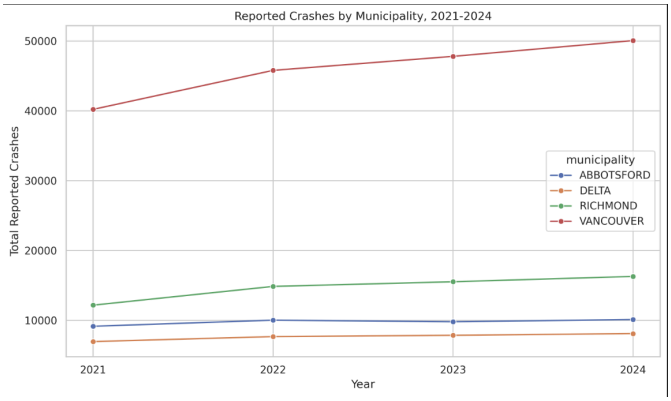


Figure 1. Annual Reported crash totals across the municipalities from 2021-2024

Despite the increase in crash counts, over the years, fortunately, victim counts did not increase at the same rate. The victims per 100 reported crashes declined across all 4 municipalities. This suggests that the increase in reported crash frequency did not correspond to an equivalent increase in casualty outcomes.

4.2 Monthly and Seasonal Patterns

Monthly patterns were surprisingly consistent across the four municipalities. February had the **lowest average reported crash count** in every municipality, while December had the **highest average crash count** in every municipality.

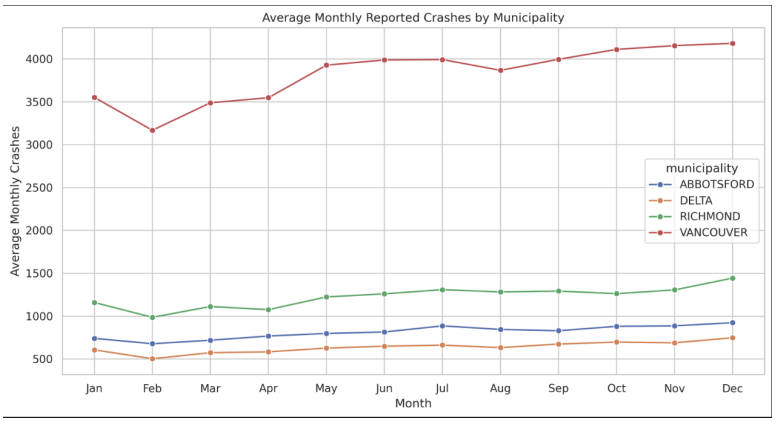
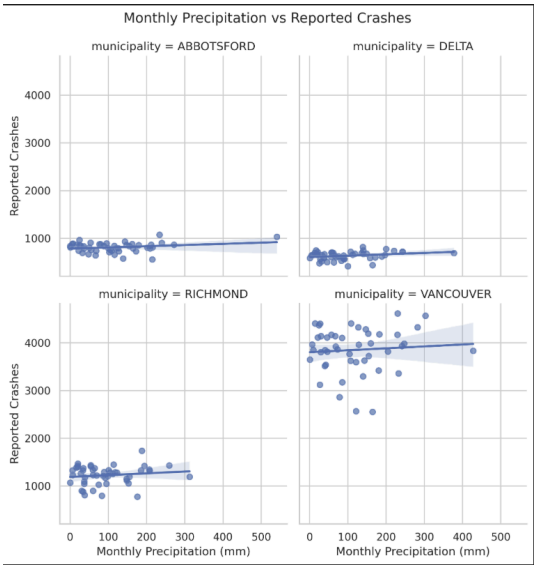


Figure 2. Average reported crash counts by calendar month and municipality, averaged across 2021–2024

When the months were grouped by season, Fall had the **highest average monthly crash count** in all four municipalities. For example, Vancouver averaged approximately 3,634 crashes per winter month, 3,654 in spring, 3,949 in summer, and **4,087 in fall**. Richmond similarly increased from approximately 1,196 per winter month to **1,287 during fall**. These results indicate an underlying seasonal structure in crash frequency. This is important when evaluating weather relationships because temperature and precipitation are also strongly seasonal.

4.3 Precipitation, Temperature, and Crash Frequency

Municipality specific correlations showed only the weak relationships between monthly precipitation and crash frequency. The precipitation Pearson correlations were approximately **0.079 for Vancouver, 0.135 for Richmond, 0.258 for Delta, and 0.239 for Abbotsford**. Although, none were statistically significant at an alpha level of 0.05. Delta had the strongest precipitation correlation, however, its P-value was approximately **0.077**, which remained above the significance threshold. Spearman correlations produced the same overall conclusion as well.



Temperature produced similarly weak results. Pearson correlations between monthly mean temperature and crashes were approximately **0.229 for Vancouver, 0.167 for Richmond, 0.084 for Delta, and 0.200 for Abbotsford**, with no P-value below 0.05. Spearman correlations were also non-significant as the closest result was Abbotsford's Spearman correlation of approximately 0.263 with a P-value of approximately 0.071.

The WET vs. DRY Mann–Whitney U tests also found no statistically significant differences. Vancouver had mean monthly crash counts of approximately **3,849 during wetter months and 3,855 during drier months** with a P-value of 0.774. Richmond's WET and DRY means were approximately 1,214 in both groups with a P-value of 0.926. Delta produced a P-value of 0.302 and Abbotsford 0.853.

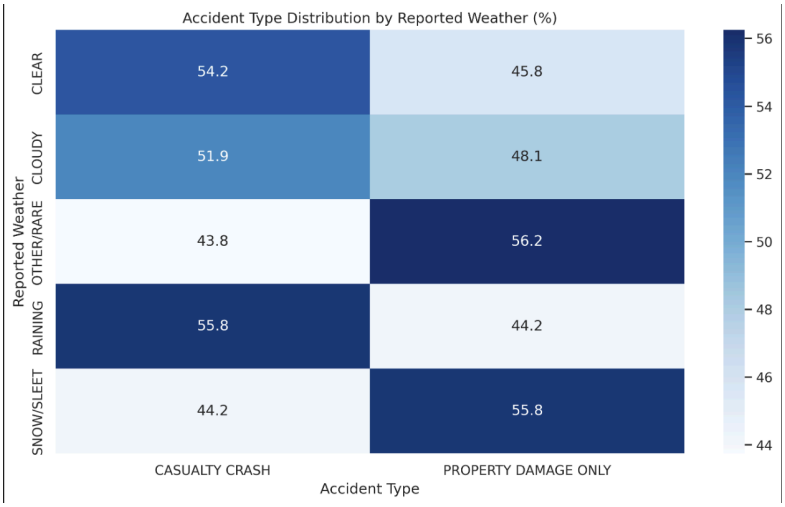
Overall, the combined evidence did **not** support a statistically significant association between the monthly precipitation, monthly mean temperature, or WET/DRY months and monthly crash frequencies within the four municipalities.

4.4 Police - Related Weather and Road Conditions

The police - reported dataset provided a second perspective because weather and road conditions were recorded in relation to individual crash categories rather than inferring them from monthly climate measurements.

Most recorded crashes occurred during **clear weather (7,766 crashes)**, followed by cloudy conditions (1,973) and rain (1,886). Most also occurred on **dry roads (8,850 crashes)**, followed by wet roads (2,814). These totals should not be interpreted as evidence that clear or dry conditions are more dangerous, because the dataset does not contain exposure information showing how often drivers encountered each condition.

Casualty rates were similar for several common conditions. Clear weather produced approximately **72.2 casualties per 100 crashes**, compared with **72.3 during rain** and 69.5 during cloudy weather. Dry and wet roads were also very similar at approximately **71.7 and 72.0 casualties per 100 crashes**, respectively. Very rare conditions such as strong wind or hail produced unstable rates because only a small number of crashes were recorded. A Chi-square test found a statistically significant association between **reported weather and accident type**, with x^2 ~ 16.55, 4 degrees of freedom, and P-value ~ 0.0024. However, Cramer's V was only approximately **0.037**, indicating that the association was weak in magnitude.



For example, approximately **54.2% of crashes recorded under clear conditions were casualty crashes**, compared with 55.8% under rain and 44.2% under snow/sleet.

Road condition and accident type were also significantly associated, with x^2 ≈ 24.17, 4 degrees of freedom, and P-value <0.001. Cramer's V was approximately **0.045**, again indicating a statistically detectable but weak association. Expected counts for both Chi-square tests were comfortably above five, satisfying the expected frequency requirement.

Limitations and Conclusion

Some limitations that came across during the project were not having traffic volume, population exposure, road design, construction activity, enforcement, or how long drivers were exposed to the elements. Since the weather was aggregated by month, this may have hidden short duration conditions such as heavy rainfall suddenly or other hail/sleet storms.

Overall, reported crash frequency increased from 2021 to 2024 in all 4 municipalities, and clear monthly and seasonal crash patterns were identified. However, the statistical tests found no significant municipality relationships between monthly precipitation or temperature, and crash frequency. These findings suggest that broad monthly weather totals alone only provide limited explanations, while conditions recorded on site show differences in accident type distributions.

Accomplish Statement

Weather and Vehicle Crash Data Analysis — CMPT 353

- **Built and validated a multi-source data pipeline** using Python and Pandas to clean and combine **309,750 ICBC crash records, 12,230 police-reported crash records, and 5,844 daily weather observations** from Environment and Climate Change Canada into 192 municipality-month observations covering four Lower Mainland municipalities from 2021–2024
- **Applied statistical analysis techniques including Pearson and Spearman correlation, Mann–Whitney U tests, multiple linear regression, and Chi-square tests** to evaluate relationships between precipitation, temperature, road conditions, and reported crash outcomes, finding no significant municipality-level monthly weather/crash association but statistically significant weak associations in police-reported categorical conditions
- **Developed reproducible data-quality validation and visual analytics** using Pandas, SciPy, scikit-learn, Matplotlib, and Seaborn, identifying a **10.6%–33.8% increase in reported crashes across the four municipalities** while documenting declining victims-per-100-crashes and resolving incomplete Richmond weather-station coverage without imputing artificial observations